PROJECT – 1

INTRODUCTION:

E-commerce furniture is a growing part of the online shopping industry. It refers to buying and selling furniture through the internet using websites or mobile apps. This project focuses on understanding how the furniture business is being transformed by e-commerce, the benefits it offers, and the challenges it faces.

In the past, people had to visit physical stores to choose and buy furniture. This process could take a lot of time and effort. Customers had to travel, look at limited stock, and often had to wait for delivery. Today, with the help of e-commerce, this has changed. Customers can now explore a wide variety of furniture from different brands, styles, and price ranges while sitting at home. They can view pictures, read product details, check customer reviews, compare prices, and even see how a product would look in their space using Augmented Reality (AR) features.

One of the biggest advantages of e-commerce furniture is convenience. People can shop 24/7 without the need to travel. It also saves time and often money, as online stores may offer discounts and deals that physical stores don't. Another benefit is variety. E-commerce platforms allow access to products from different regions or even different countries, offering more options than a local store.

Another challenge is logistics. Delivering large furniture items safely and quickly can be difficult and costly. Some companies partner with delivery services or have their own logistics systems to manage this better. Some also offer free installation and setup to improve customer satisfaction.

Many well-known brands like IKEA, Pepper fry, Urban Ladder, and Wayfair have made big success in e-commerce furniture. Small and local sellers are also using online platforms like Amazon, Flipkart, or even their own websites to sell furniture.

However, there are some challenges too. One issue is that furniture is a large and often expensive item. Customers may worry about the quality, comfort, or size of the product since they can't touch or test it in person. There is also the risk of receiving damaged goods during delivery. To solve this, many online furniture companies offer clear return policies, detailed product descriptions, 3D images, and customer support.

the e-commerce furniture market is growing fast. It offers a better and more flexible way for people to shop for furniture and helps businesses expand. With the use of technology and good customer service.

DATASET OVERVIEW:

This dataset comprises 2,000 entries scraped from AliExpress, detailing a variety of furniture products. It captures key sales metrics and product details through 5 main columns: productTitle, originalPrice, price, sold, and tagText. These features provide insight into product pricing, popularity, and categorization. The dataset offers a snapshot of consumer purchasing patterns and market trends in the online furniture retail space.

APPLICATIONS:

The dataset is ripe for exploratory data analysis, market trend identification, and price optimization studies. It also supports predictive modelling to forecast sales, evaluate the impact of discounts on purchase volume, and uncover relationships between product features and popularity. These insights can guide strategic decision-making in e-commerce and digital retail environments.

TECHNOLOGY [TOOLS]:

PYTHON:

Python is a high-level, easy-to-learn programming language that is widely used around the world. It was created by **Guido van Rossum** and was first released in **1991**. Python is popular because of its simple and readable syntax, which makes it a great choice for beginners and professionals alike.

Python can be used for many purposes, such as **web development, data analysis, artificial intelligence, machine learning, automation, game development**, and more. It is a powerful and flexible language that helps programmers build a wide variety of applications quickly and efficiently.

One of the main reasons for Python's popularity is its large collection of libraries and frameworks, such as **NumPy** for mathematics, **Pandas** for data analysis, **Django** and **Flask** for web development, and **TensorFlow** for machine learning.

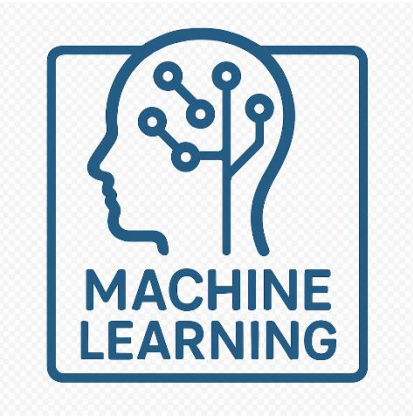
MEACHIN LEARNING:

Machine Learning (ML) is a branch of artificial intelligence (AI) that allows computers to learn from data and make decisions or predictions without being directly programmed. Instead of writing rules for every situation, we give the computer lots of data, and it learns patterns from that data to perform tasks on its own.

For example, machine learning is used in real life for things like recognizing faces in photos, suggesting videos on YouTube, predicting the weather, or even detecting fraud in banking.

There are different types of machine learning:

- Supervised learning, where the machine is trained on labeled data (data with correct answers).
- Unsupervised learning, where the machine finds patterns in data without labels.
- Reinforcement learning, where the machine learns by trying actions and getting rewards or penalties.



Machine learning uses tools like statistics, algorithms, and programming (often in languages like Python) to process and learn from large amounts of data. It is widely used in fields like healthcare, finance, transportation, and entertainment.

STRUCTURED QUERY LANGUAGE [SQL]:

SQL stands for Structured Query Language. It is a special programming language used to manage and work with data stored in relational databases. SQL helps users to create, read, update, and delete data — this is often referred to as CRUD operations.

Some of the most common SQL commands are:

- SELECT – to retrieve data from a database
- INSERT – to add new data
- UPDATE – to change existing data
- DELETE – to remove data
- CREATE – to make new tables or databases

- DROP – to delete tables or databases

SQL is used in many areas such as **web development, data analysis, business intelligence, and software development**. It works with many popular database systems like **MySQL, PostgreSQL, Oracle, Microsoft SQL Server**, and **SQLite**.

MICROSOFT EXCEL:

Microsoft Excel is a powerful spreadsheet program developed by Microsoft. It is widely used in homes, schools, and businesses for organizing, analyzing, and storing data. Excel allows users to create and manage data in a grid of rows and columns, known as a worksheet.

One of the main features of Excel is its ability to perform **calculations and mathematical functions**. You can use formulas to quickly add, subtract, multiply, divide, and do much more. Excel also offers many built-in functions for handling data, such as **SUM, AVERAGE, IF, VLOOKUP**, and more.

Another powerful tool in Excel is **PivotTables**, which allow users to summarize large amounts of data quickly. Excel can also work with other software and support data import/export from various sources.

In simple words, Excel is a useful tool for managing numbers, keeping records, making reports, and analyzing information in a smart and organized way.

E-COMMERCE FURNITURE

COLUMN DESCRIPTIONS:

This dataset contains information about various products, with each entry including several key attributes,

- productTitle: - The product title columns hold the name or title of the product, which helps identify it in listing or catalogs.

- originalPrice: - The original price represents the initial price set for the product before any discounts or promotions.

- price: - The price column indicates the current selling price, reflecting any applied discounts.
- sold: - The sold column shows the number of units that have been purchased, giving insight into the product popularity or sales performance.

- tagText: - The tag text column contains special labels or tags, such as promotional highlights like “Free shipping” or “Shipping charge”.

Ethically Collected Data:

The data was collected in accordance with ethical data gathering practices, ensuring full compliance with AliExpress’s terms of service and respect for user privacy. Only publicly available product information was scraped, with no personal or sensitive user data involved. This approach aligns with ethical standards for data usage in research and analytics.

Acknowledgements:

This dataset was created using data sourced from AliExpress, with the help of Apify for web scraping automation. The thumbnail features alongside this dataset was generously provided by Spacejoy on Unsplash. We extend our sincere thanks to these platforms for making this project possible.

Photo by Spacejoy on Unsplash.

E-COMMERCE FURNITURE

Project Overview:

- **Objective: -**
  To analyse online furniture sakes data from AliExpress in order to uncover pricing trends, consumer purchasing behaviour, and the impact of discount – ultimately enabling data-driven insights for sales forecasting and market strategy.

- **Tech Stack: -**
  Python, pandas, scikit-lean, matplotlib, seaborn

Steps:

1. Data Collection
2. Data Preprocessing
3. Feature Engineering
4. Exploratory Data Analysis (EDA)
5. Model Selection and Training
6. Model Evolution
7. Concussion

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1. Data Collection

In this step, the dataset is available in csv format. We can load it using pandas.

```
# Import necessary libraries

import pandas as pd


# Load dataset

df = pd.read_csv('ecommerce_furniture_dataset_2024.csv')


# View the first few rows of the dataset

print(df.head())
```

2. Data Preprocessing

We will clean the data by handling missing values, converting categorical variables, and removing irrelevant columns.

```
# Check for missing values

print(df.isnull().sum())

# Dropping any rows with missing values (if necessary)

df = df.dropna()

# Coverting tagText into a categorical features (if necessary)

Df['tagText'] = df['tagText'].astype('category').cat.codes

# Checking for data types and conversions (if necessary)

print(df.info())
```

3. Exploratory Data Analysis (EDA)

Visualize the relationships between features and the target variable (sold) Understand the distribution and trends in the data.

```
import seaborn as sns

import matplotlib.pyplot as plt


# Distribution of 'sold' values

sns.histplot(df['sold'], kde=True)

plt.title('Distribution of Furniture items Sold')

plt.show()


# Plot the relationship between originalPrice, price and sold

sns.pairplot(df, vars=['originalPrice', 'price','sold'], kind='scatter'
plt.title('Relationship Between Price, Original Price, and Item sold')

plt.show()
```

4. Feature Engineering

1. **Handling product Titles:** We will convert productTitle to numeric features using techeniques like TF-IDF.
2. **Price and Discount Features:** Create a new feature to calculate the percentage discount from originalPrice and price.

```
from sklearn.feature_extraction.text
import TfidfVectorizer
# Creating a new feature: percentage discount
df['discount_percentage']=((df['originalPrice']-df['price']) / df['originalPrice']) * 100

# Convert productTitle into a numeric feature using TF-IDFVectorizer
Tfidf = TfidfVectorizer(max_feature=100)
productTitle_tfidf = tfidf.fit_transform(df['productTitle'])

# Convert to DataFrame and concatenate to original df
productTitle_tfidf_df =
pd.DataFrame(productTitle_tfidf.toarray(),columns=tfidf.get_feature_name_out())
df = pd.concat([df.productTitle_tfidf_df], axis=1)

# Drop original productTitle as it's now encoded
df = pd.drop('productTitle', axis=1)
```

5. Model Selection & Training

We will use **Linear Regression** and **Random Forest Regression** as models to predict the number of items sold.

```
from sklearn.model_selection import train_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score

# Split the dataset into features (X) and target (y)
X = df.drop('sold', axis=1)
y = df[ 'sold' ]

# Train-test split (80% train, 20 test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Initialize models
lr_model = LinearRegression()
rf_model = RandomForestRegressor(n_estimators=100, random_state=42)
```

```
# Train models
lr_model.fit(X_train, y_train)
rf_model.fit(X_train, y_train)
```

6. Model Evolution

We evaluate the model's performance using **mean squared error (MSE)** and R-squared metrics.

```
# Predict with Linear Regression
y_pred_lr = lr_model.predict(X_test)
mse_lr = mean_squared_error(y_test, y_pred_lr)
r2_lr = r2_score(y_test, y_pred_lr)

# Predict with Random Forest
y_pred_rf = rf_model.predict(X_test)
mse_rf = mean_squared_error(y_test, y_pred_rf)
r2_rf = r2_score(y_test, y_pred_rf)

# print model evaluation results

print(f'Linear Regression MSE: {mse_lr}, R2: {r2_lr}')
print(f'Random Forest MSE: {mse_rf}, R2: {r2_rf}')
```

7. Concussion

After evaluating the models, we can conclude which model performed better and further tune hyper parameters if needed. Random Forest tends to perform better on complex datasets with high variance, while Linear Regression might work well if relationships are linear.

Output: -

Linear Regression Model: MSE and R-squared score

Random Forest Model: MSE and R-squared score

PROJECT CODE

In [1]:

```
# Import necessary libraries
import numpy as np
import pandas as pd

import os
for dirname, _, filenames in os.walk('/content'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

OUTPUT:

```
/content/ecommerce_furniture_dataset_2024.csv
```

In [2]:

```
# Lord dataset from csv file
df = pd.read_csv
('/content/ecommerce_furniture_dataset_2024.csv',
encoding='latin1')
```

In [3]:

```
# View the first few rows of the dataset
df.head()
```

OUTPUT [3]:

	productTitle	originalPrice	price	sold	tagText
0	Dresser For Bedroom With 9 Fabric Drawers Ward...	NaN	$46.79	600	Free Shipping
1	Outdoor Conversation Set 4 Pieces Patio Furnit...	NaN	$169.72	0	Free Shipping

2	Dresser For Bedroom With 7 Fabric Drawers Orga...	$78.40	$39.46	7	Free Shipping
3	Modern Accent Boucle Chair, Upholstered Tufted...	NaN	$111.99	0	Free Shipping
4	Small Unit Simple Computer Desk Household Wood...	$48.82	$21.37	1	Free Shipping

In [4]:

```
# Check for missing values
print(df.isnull().sum())
```

OUTPUT [4]:

```
productTitle        0
originalPrice    1536
price               6
sold                1
tagText             1

dtype: int64
```

In [5]:

```
df.shape
```

OUTPUT [5]:

```
(2000, 5)
```

In [6]:

```
# Dropping any rows with missing values (if applicable)
df.drop(['originalPrice'],axis=1,inplace=True)
```

In [7]:

```
df.head()
```

OUTPUT [7]:

	productTitle	price	sold	tagText
0	Dresser For Bedroom With 9 Fabric Drawers Ward...	$46.79	600	Free Shipping
1	Outdoor Conversation Set 4 Pieces Patio Furnit...	$169.72	0	Free Shipping
2	Dresser For Bedroom With 7 Fabric Drawers Orga...	$39.46	7	Free Shipping
3	Modern Accent Boucle Chair, Upholstered Tufted...	$111.99	0	Free Shipping
4	Small Unit Simple Computer Desk Household Wood...	$21.37	1	Free Shipping

In[8]:

```
df['tagText'].nunique()
```

OUTPUT [8]:

96

In [9]:

```
df['tagText'].value_counts()
```

OUTPUT [9]:

```
tagText                          count

Free Shipping                    1886
+Shipping: $5.09                 9
+Shipping: $239.64               2
+Shipping: $80.21                2
+Shipping: $94.92                2

                - - -

 +Shipping: $134.27              1
+Shipping: $151.69               1
+Shipping: $41.93                1
+Shipping: $78.61                1
+Shipping: $171.49               1

96 rows x 1 columns
dtype: int64
```

In [10]:

```
df['tagText'] = df['tagText'].apply(lambda x: x if x in ['Free
Shipping', '+Shipping: $5.09'] else 'others')

# Display the modified value counts
print((df['tagText'].value_counts()))
```

OUTPUT [10]:

```
tagText
Free Shipping    1886
others            114

Name: count, dtype: int64
```

In [11]:

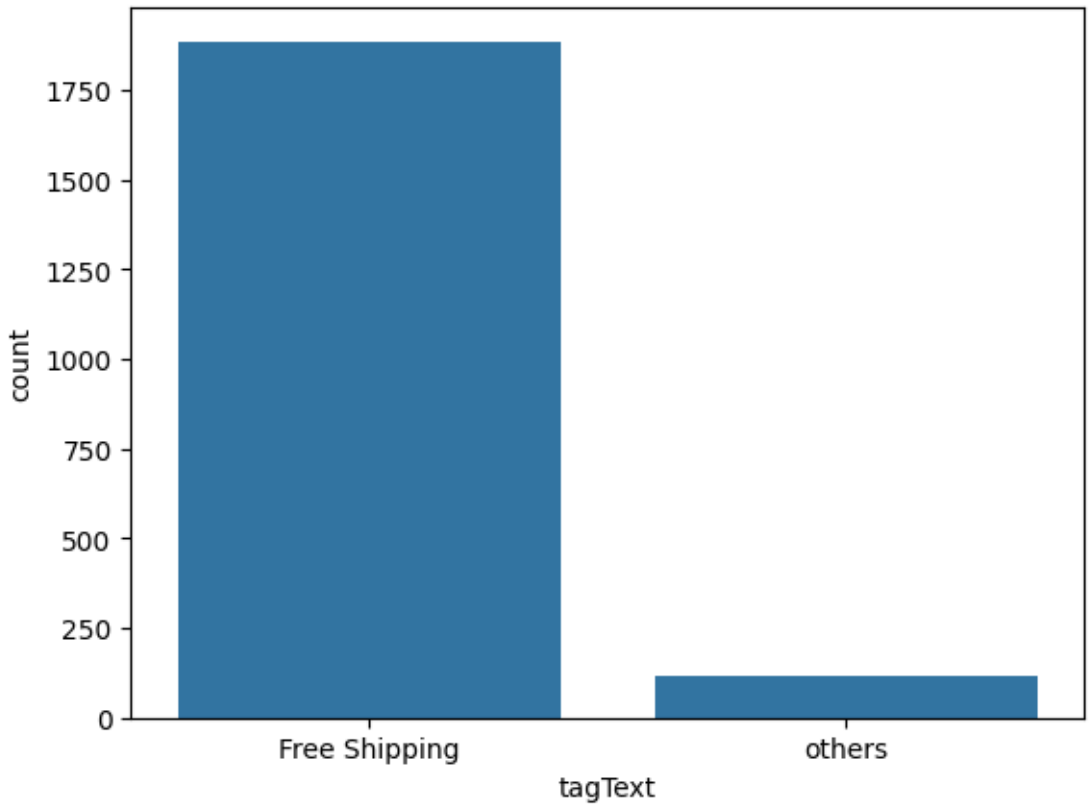
```
# importing another libraries
import seaborn as sns
```

In [12]:

```
sns.countplot(x = 'tagText', data = df)
```

OUTPUT [12]:

```
<Axes: xlabel='tagText', ylabel='count'>
```



In [13]:

```
# Remove $ and split by tab (\t), then take the first part as
the price
df['price_clean'] = df['price'].str.replace('$', '',
regex=False).str.split('\t').str[0]

# Convert the cleaned price to float
df['price_clean'] = df['price_clean'].astype(float)
```

In [14]:

```
df['price'] = df['price'].str.replace('$', '',
regex=False).str.split('\t').str[0].astype(float)
```

In [15]:

```
df.drop(['price_clean'],axis=1,inplace=True)
```

In [16]:

```
df.head()
```

OUTPUT [16]:

	productTitle	price	sold	tagText
0	Dresser For Bedroom With 9 Fabric Drawers Ward...	46.79	600	Free Shipping
1	Outdoor Conversation Set 4 Pieces Patio Furnit...	169.72	0	Free Shipping
2	Dresser For Bedroom With 7 Fabric Drawers Orga...	39.46	7	Free Shipping
3	Modern Accent Boucle Chair, Upholstered Tufted...	111.99	0	Free Shipping
4	Small Unit Simple Computer Desk Household Wood...	21.37	1	Free Shipping

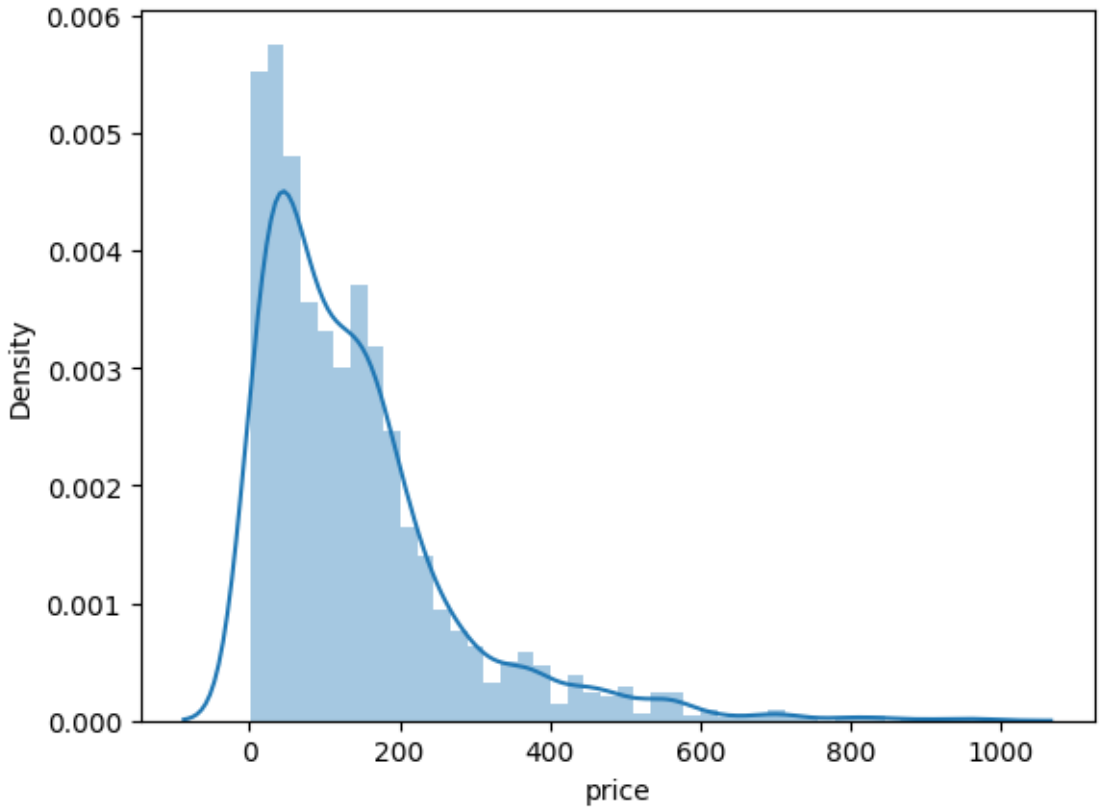
In [17]:

```
sns.distplot(df['price'])
```

```
<ipython-input-35-86c1ddc3c66a>:1: UserWarning:

`distplot` is a deprecated function and will be removed in
seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level
function with
similar flexibility) or `histplot` (an axes-level function for
histograms).

For a guide to updating your code to use the new functions,
please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe57
51

  sns.distplot(df['price'])
```

OUTPUT [17]:

```
<Axes: xlabel='price', ylabel='Density'>
```



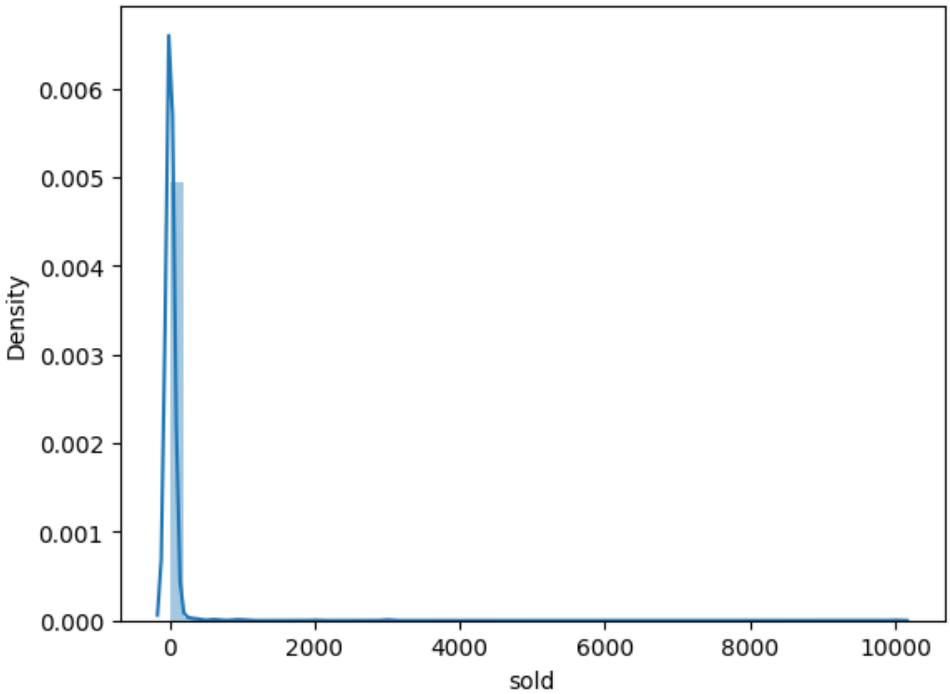
In [18]:

```
sns.distplot(df['sold'])
```

```
<ipython-input-36-17cdf52c425f>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn
v0.14.0.

Please adapt your code to use either `displot` (a figure-level function
with
similar flexibility) or `histplot` (an axes-level function for
histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['sold'])
```

OUTPUT [18]:

```
<Axes: xlabel='sold', ylabel='Density'>
```

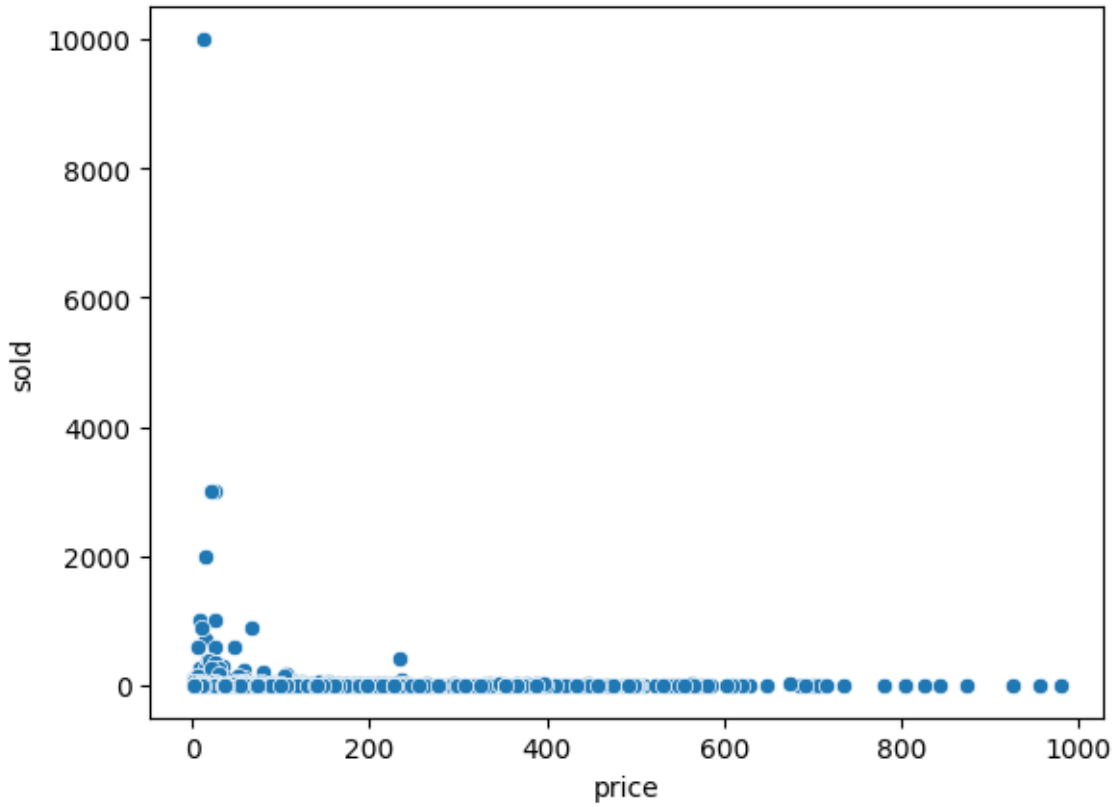


In [19]:

```
sns.scatterplot(x='price', y='sold', data=df)
```

OUTPUT [19]:

```
<Axes: xlabel='price', ylabel='sold'>
```

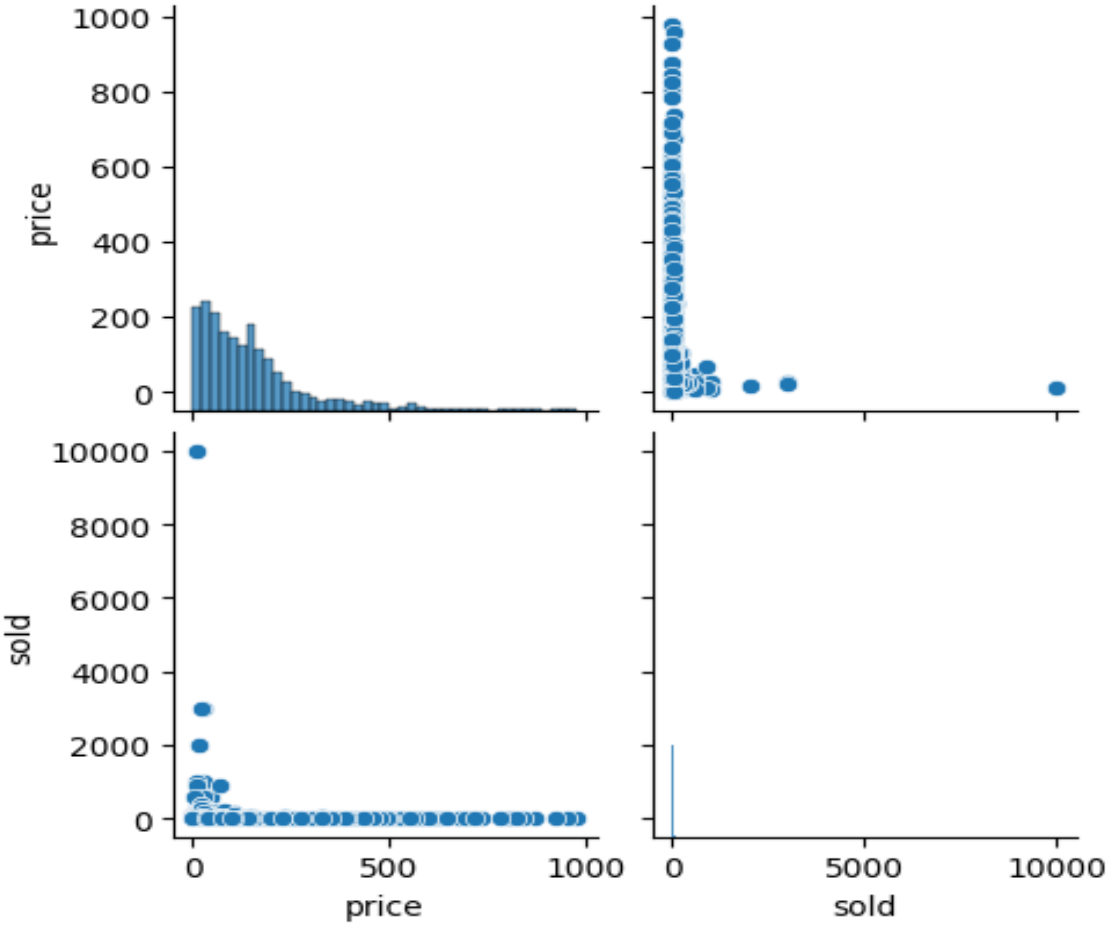


In [20]:

```
filtered_df = df[df['tagText'] == 'Free Shipping']

# Create a pairplot including the 'sold' column and other
relevent column
sns.pairplot(filtered_df[['price', 'sold']])
```

OUTPUT [20]:

```
<seaborn.axisgrid.PairGrid at 0x79fb5bc8fe90>
```



In [21]:

```
from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
df['tagText']=le.fit_transform(df['tagText'])
```

In [22]:

```
df.head()
```

OUTPUT [22]:

	productTitle	price	sold	tagText
0	Dresser For Bedroom With 9 Fabric Drawers Ward...	46.79	600	0
1	Outdoor Conversation Set 4 Pieces Patio Furnit...	169.72	0	0
2	Dresser For Bedroom With 7 Fabric Drawers Orga...	39.46	7	0
3	Modern Accent Boucle Chair, Upholstered Tufted...	111.99	0	0
4	Small Unit Simple Computer Desk Household Wood...	21.37	1	0

In [23]:

```
df['tagText'].value_counts()
```

OUTPUT [23]:

```
tagText              count

0                    1886
1                    114

Name: count, dtype: int64
```